

NAME SCHOOL TEACHER 

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Pre-Junior Certificate Examination, 2018

# Mathematics

Paper 2

Higher Level

Time: 2 hours, 30 minutes

300 marks

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| School Stamp |
|              |

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| Running Total |  |
|---------------|--|

| For Examiner |      |          |      |
|--------------|------|----------|------|
| Question     | Mark | Question | Mark |
| 1            |      | 11       |      |
| 2            |      |          |      |
| 3            |      |          |      |
| 4            |      |          |      |
| 5            |      |          |      |
| 6            |      |          |      |
| 7            |      |          |      |
| 8            |      |          |      |
| 9            |      |          |      |
| 10           |      | Total    |      |

|       |
|-------|
| Grade |
|       |

## Instructions

There are 11 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if you do not show all necessary work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

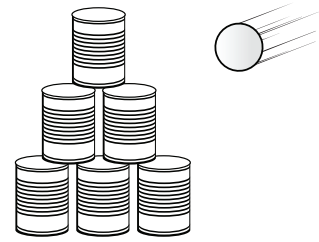
Write the make and model of your calculator(s) here:

### Question 1

(Suggested maximum time: 10 minutes)

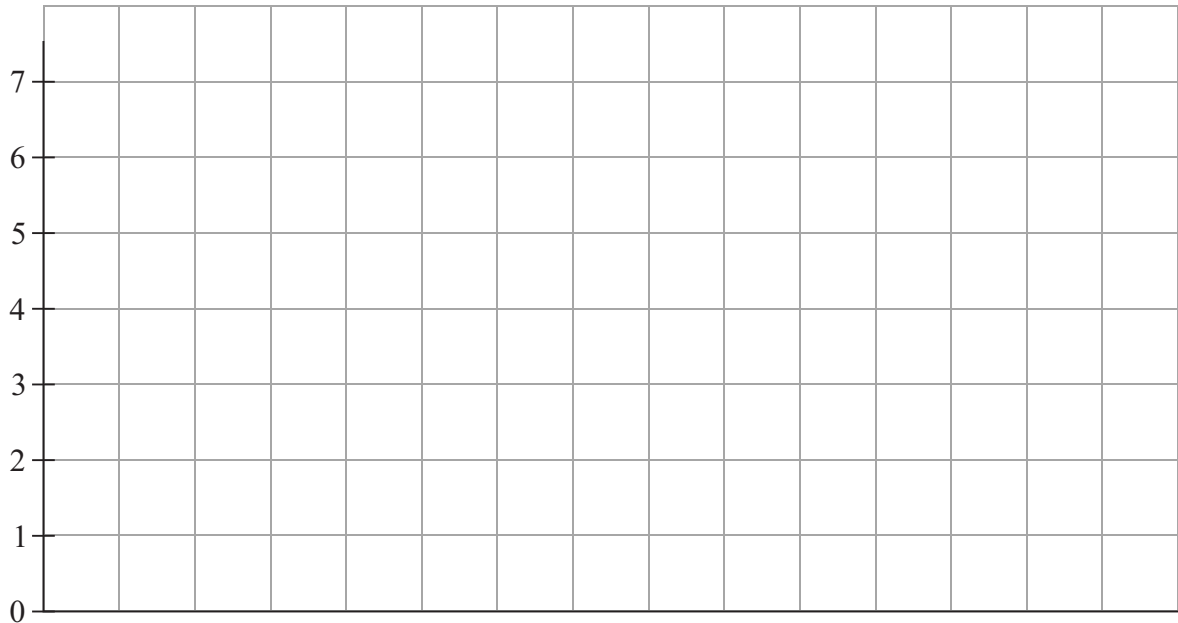
A game at a funfair involves each player throwing balls at 6 cans stacked on top of each other as shown in the picture.

The table below shows the number of cans knocked down by each player who took part in the game over an hour on a particular evening.



|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 0 | 2 | 3 | 4 | 4 | 5 | 5 | 5 |
| 5 | 5 | 5 | 5 | 6 | 6 | 6 | 6 |

(a) Draw a line plot of the data displayed above.



(b) Find the value of each of the following statistics for this data:

(i) the **mode**

(ii) the **median**

|        |          |
|--------|----------|
| Mode = | Median = |
|--------|----------|

(c) Find the **interquartile range** of this data.

|  |
|--|
|  |
|  |
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(d) Explain what the interquartile range of the data means in this context.

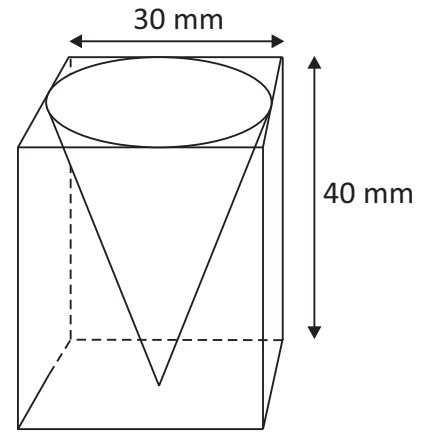
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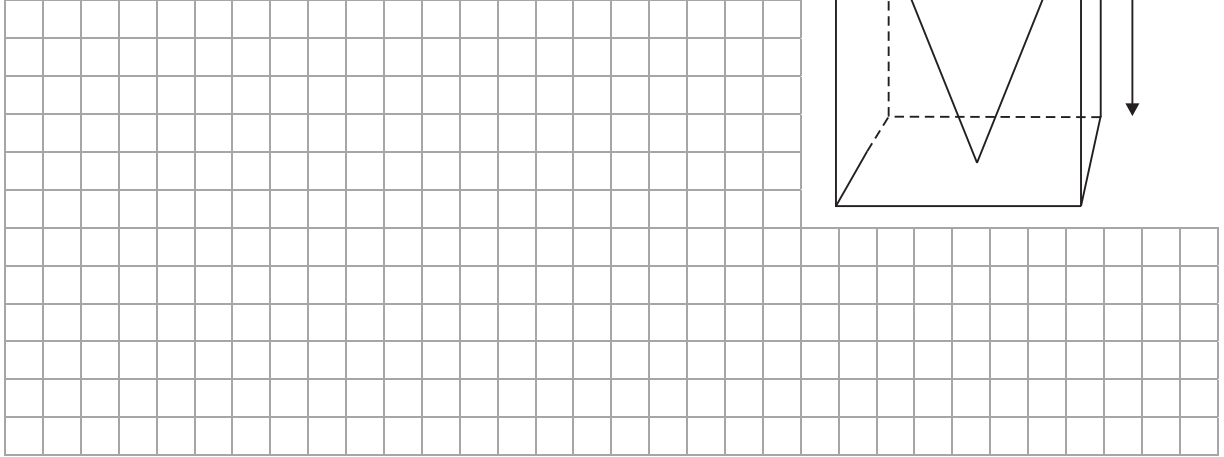
## Question 2

(Suggested maximum time: 20 minutes)

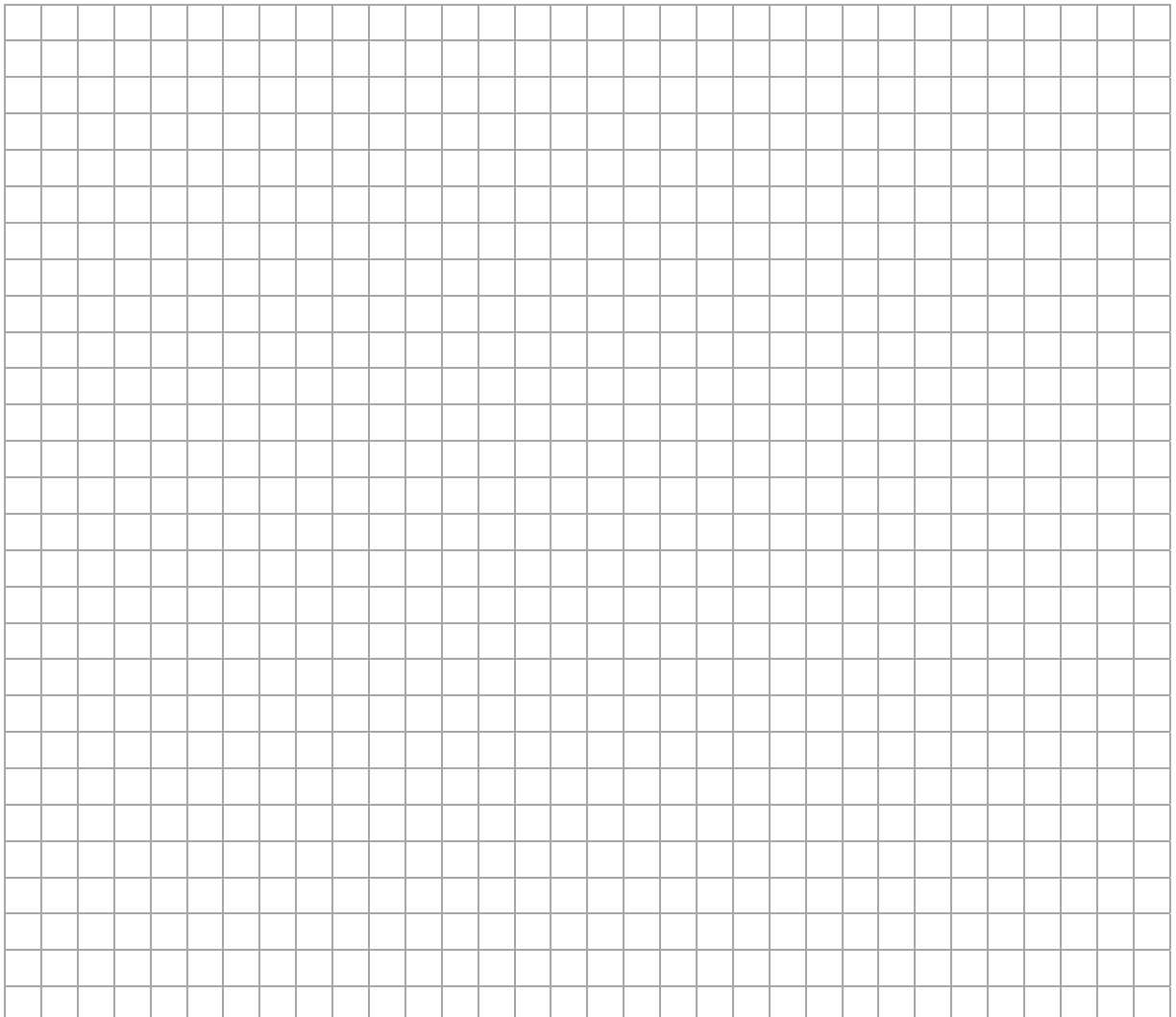
A cone of diameter 30 mm and height 40 mm sits exactly inside a closed box in the shape of a cuboid, as shown in the diagram.



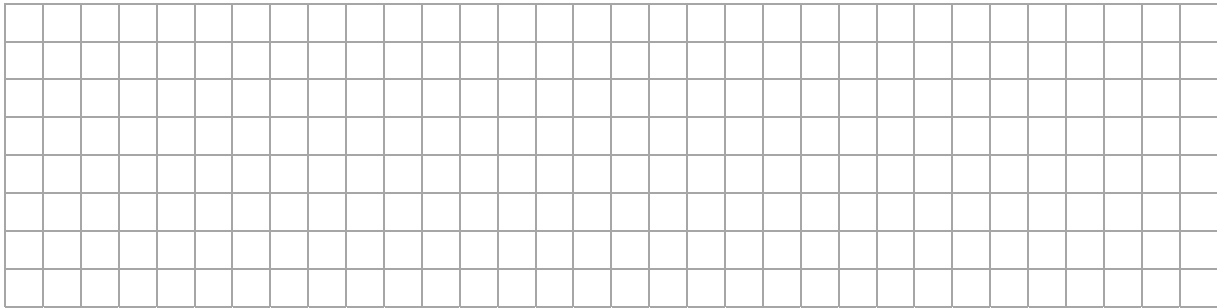
- (a) Find the **volume** of the cone.  
Give your answer in  $\text{mm}^3$  in terms of  $\pi$ .



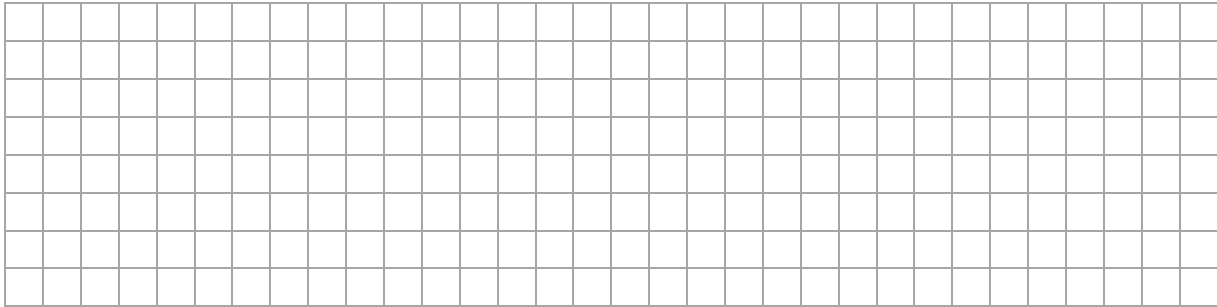
- (b) Construct an accurate **net** of the box.  
Show all of your construction lines clearly.



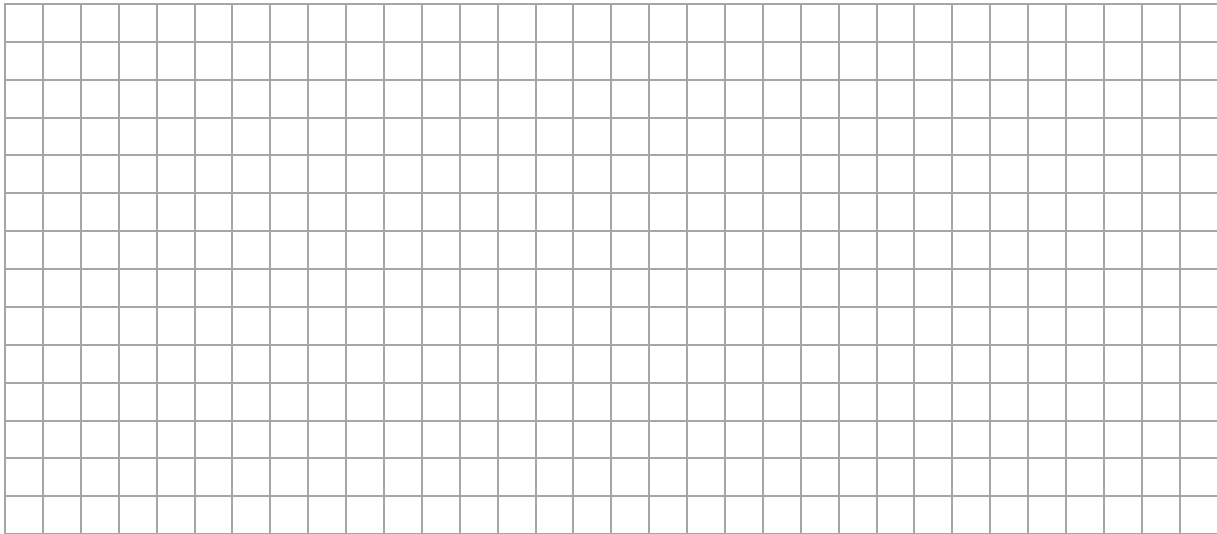
(c) Calculate the **surface area** of the box.



(d) Calculate the **volume** of the box.

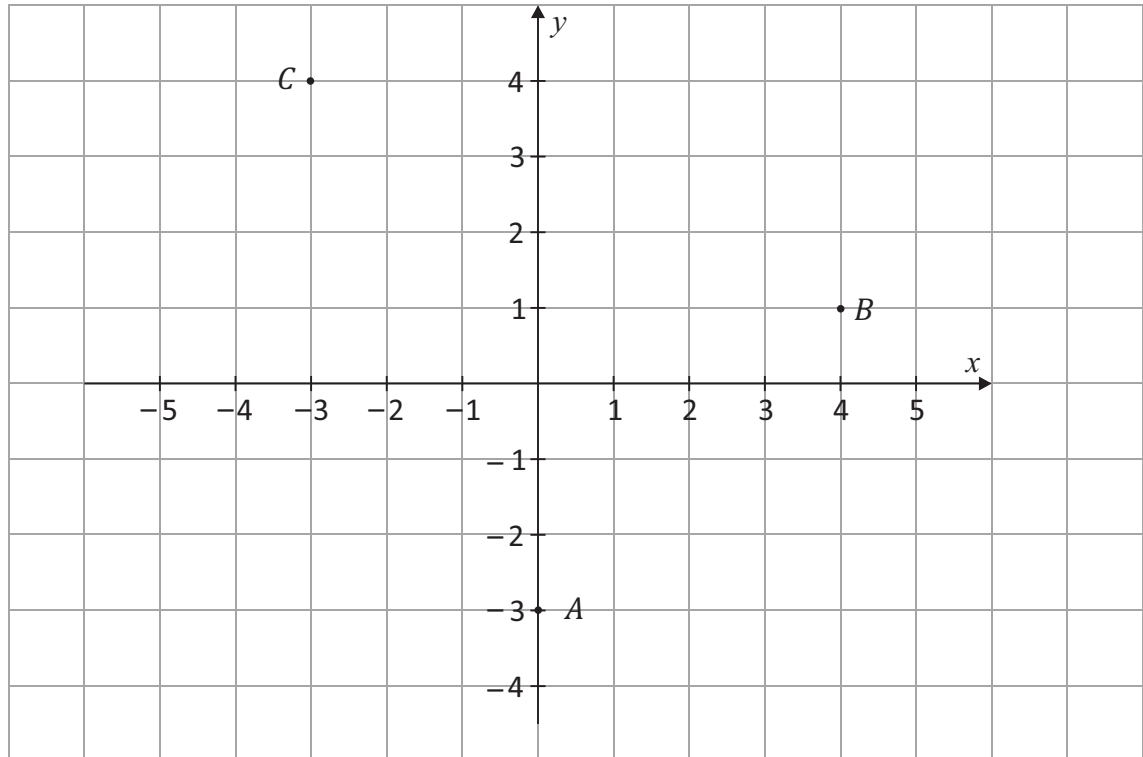


(e) A cylinder of height 20 mm has the same volume as the box.  
Find the **radius** of this cylinder. Give your answer correct to the nearest millimetre.



**Question 3****(Suggested maximum time: 20 minutes)**

The co-ordinate diagram below shows the points  $A$ ,  $B$  and  $C$ .



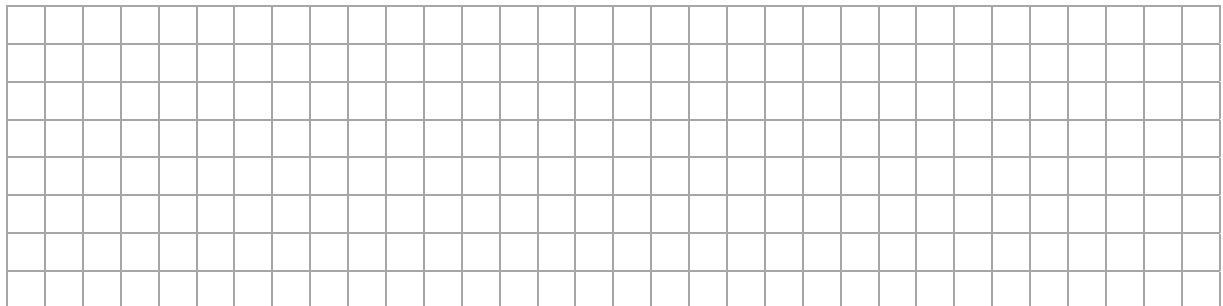
- (a)** Write down the co-ordinates of  $A$ ,  $B$  and  $C$ .

|      |  |   |  |     |
|------|--|---|--|-----|
| $A($ |  | , |  | $)$ |
|------|--|---|--|-----|

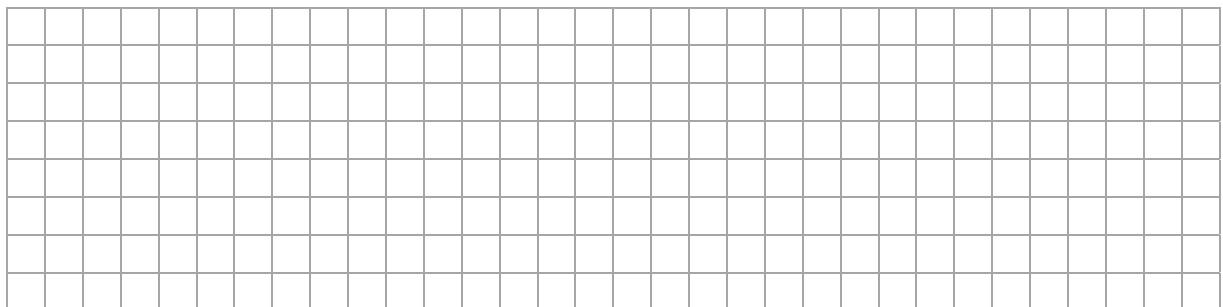
|      |  |   |  |     |
|------|--|---|--|-----|
| $B($ |  | , |  | $)$ |
|------|--|---|--|-----|

|      |  |   |  |     |
|------|--|---|--|-----|
| $C($ |  | , |  | $)$ |
|------|--|---|--|-----|

- (b)** Find the co-ordinates of  $M$ , the midpoint of  $AB$ .

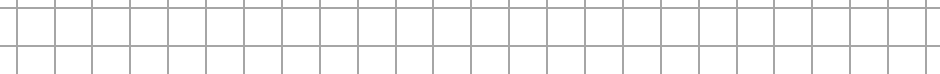


- (c)** Find the equation of the line  $AB$ .



- (d)** The line  $k$  passes through  $C$  and is perpendicular to the line  $AB$ . Find the equation of the line  $k$  in the form  $ax + by + c = 0$ .

- (e)** Investigate whether the midpoint of the line  $AB$  is also a point on the line  $k$ .



- (f)** Using the information above, state whether or not the triangle  $ABC$  is isosceles. Give a reason for your answer.

|         |  |
|---------|--|
| Answer: |  |
| Reason: |  |
|         |  |
|         |  |
|         |  |
|         |  |

- (g) The triangle  $AMC$  is the image of the triangle  $MBC$  under a transformation. Write down this transformation. Give a reason for your answer.

**Answer:**

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**Reason:**

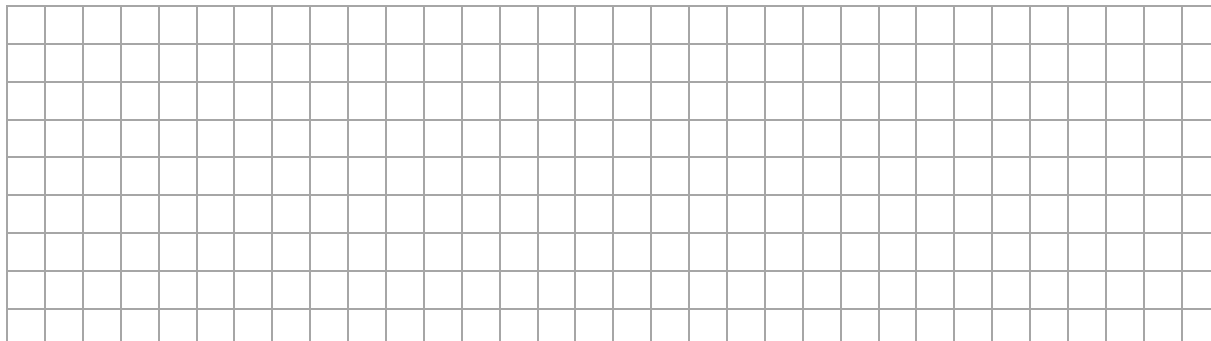
#### Question 4

(Suggested maximum time: 20 minutes)

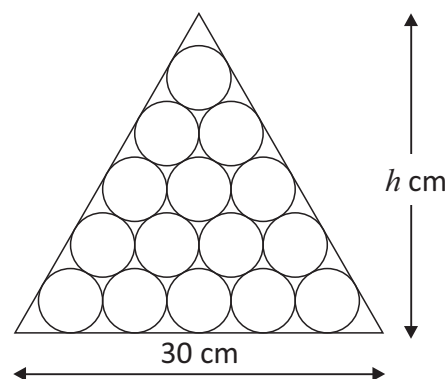
The playing surface of a standard full-size snooker table is approximately 11.7 feet by 5.8 feet.



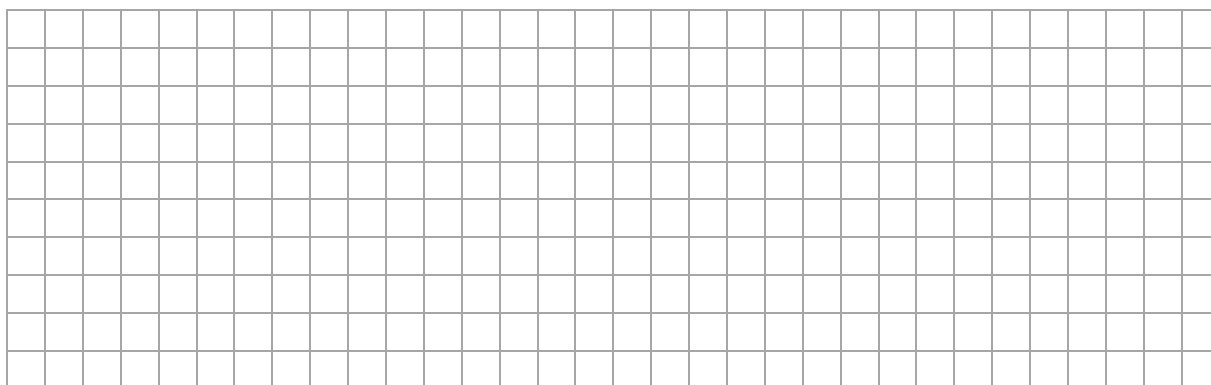
- (a) Given that 1 metre = 3.3 feet, find the area of the playing surface of a standard full-size snooker table. Give your answer in  $\text{m}^2$ , correct to three significant figures.



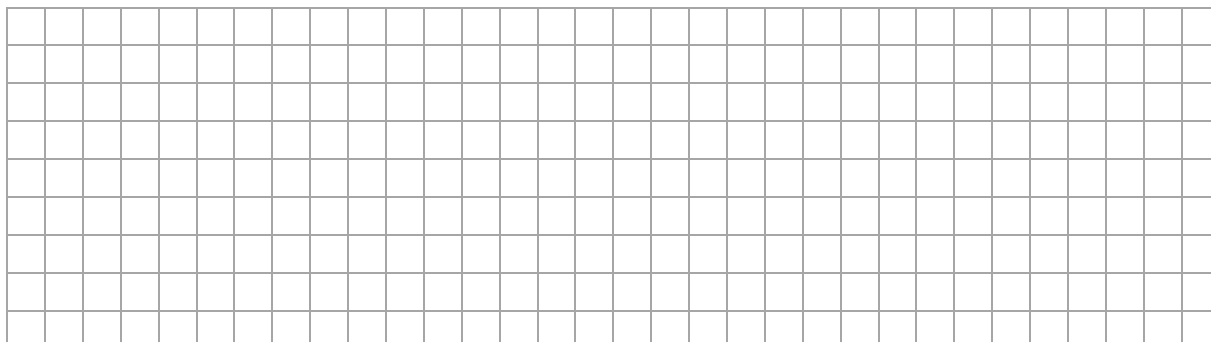
- (b) An equilateral triangular frame, with sides of length 30 cm, is used to set out the 15 red balls at the beginning of a game of snooker, as shown in the diagram.



- (i) Using the theorem of Pythagoras, or otherwise, calculate the value of  $h$  in the diagram.

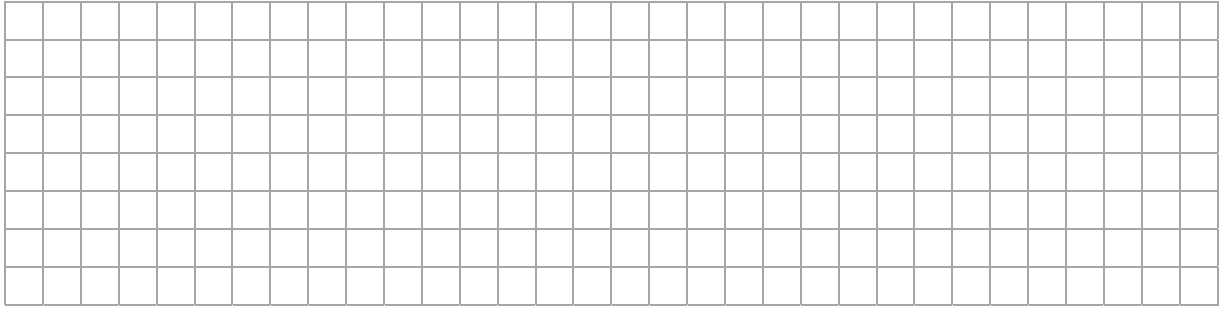


- (ii) Hence, or otherwise, find the area of the triangular frame. Give your answer in  $\text{cm}^2$ , correct to one decimal place.

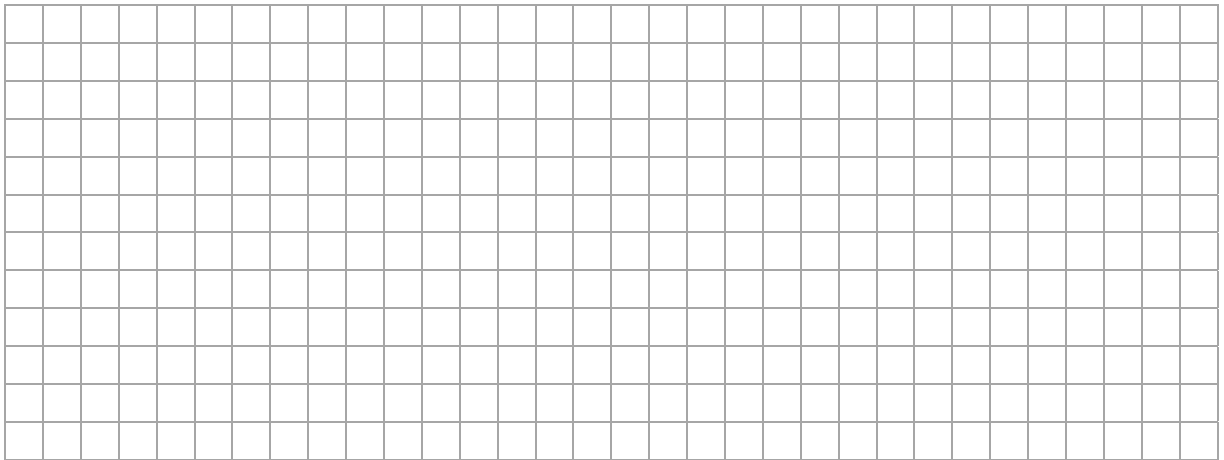
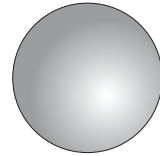




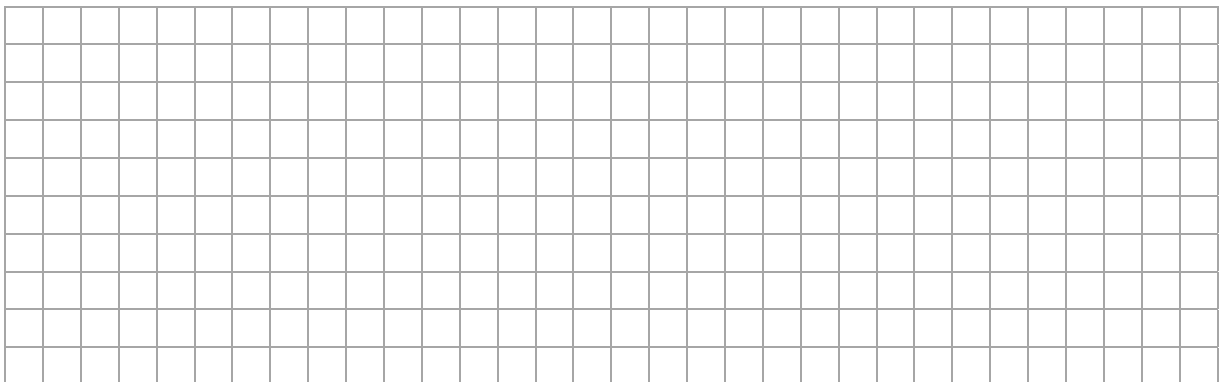
- (c) Prove, using trigonometric ratios, that an angle inside the triangular frame is  $60^\circ$ .



- (d) A standard snooker ball is in the shape of a sphere of diameter 5.25 cm.  
Find the volume of all 15 red balls.  
Give your answer in  $\text{cm}^3$ , correct to one decimal place.



- (e) Assuming that the triangular frame is the same height as the red snooker balls, express the volume of the 15 balls as a percentage of the internal volume of the triangular frame.  
Give your answer correct to one decimal place.



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### Question 5

**(Suggested maximum time: 10 minutes)**

Two ships,  $A$  and  $B$ , leave the same port,  $P$ , at 12:00, travelling perpendicularly to one another. Their journeys are mapped on the diagram.



- (a)** Ship A travels at a constant speed of 6 km/hour.  
How far had Ship A travelled by 2:20 p.m. that same day?

[illegible]

- (b)** At 2:20 p.m., Ship *B* had travelled 21 km.  
Find the average speed of Ship *B*.

[illegible]

- (c) Find the distance between Ship *A* and Ship *B* at 2:20 p.m. Give your answer in the form  $a\sqrt{b}$ , where  $a, b \in \mathbb{N}$ .

[illegible]

**Question 6** (Suggested maximum time: 10 minutes)

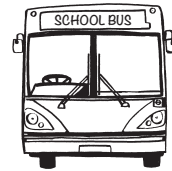
**Question 6** (Suggested maximum time: 10 minutes)

Each morning Luke and Leia need to take the same bus to school.

The probability that Luke catches the bus is 0.8, while the

probability that Leia catches it is 0.9.

The two probabilities are independent of each other.



- (a)** What is meant by the term **independent of each other** in the context of this question?

[illegible]

- (b)** What is the probability that, on a given day, both Luke and Leia catch the bus?

[illegible]

- (c)** What is the probability that, on a given day, Leia catches the bus but Luke does not?

[illegible]

- (d)** What is the probability that, on a given day, **neither** Luke nor Leia catches the bus?

[illegible]

- (e) What is the probability that, on a given day, **at least one** of them catches the bus?

[illegible]

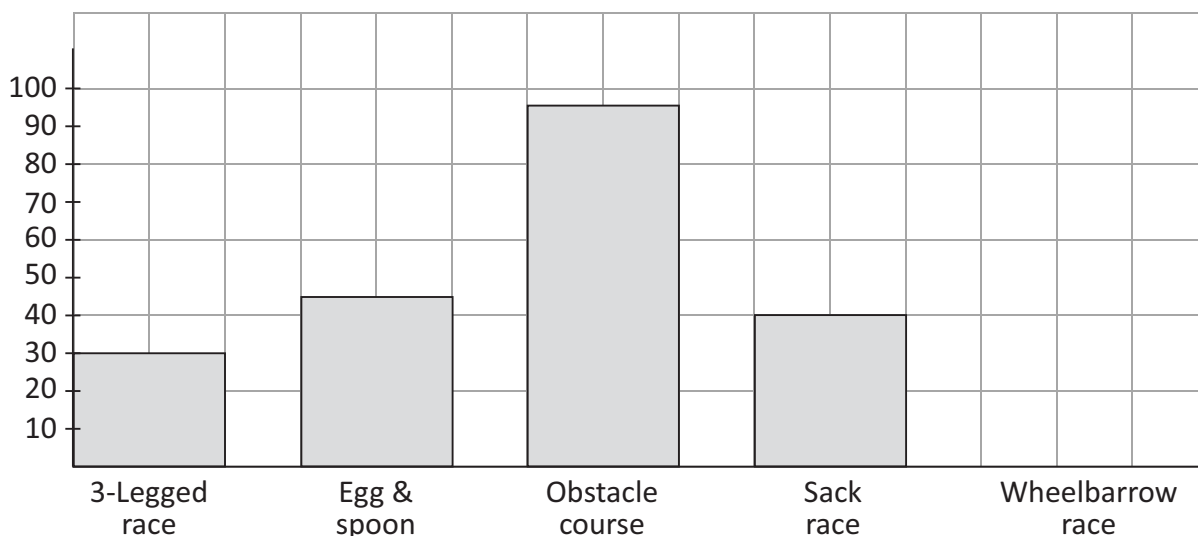
**Question 7** (Suggested maximum time: 20 minutes)

**Question 7** (Suggested maximum time: 20 minutes)

250 first and third-year students were asked which of the following novelty races they would like to participate in during their school's Sports Day.

3-legged race, egg & spoon race, obstacle course, sack race or wheelbarrow race.

The results are displayed in the following incomplete bar chart.



- (a)** What percentage of students chose the *egg & spoon* race?

[illegible]

- (b)** 36% of the students surveyed said they would like to participate in the *wheelbarrow* race. Indicate this on the bar chart.

[illegible]

- (c)** Did students choose to participate in more than one novelty race? Give a reason for your answer.

[illegible]

- (d)** Is this survey a fair sample of the school's student population? Give a reason for your answer.

[illegible]



### Question 8

**(Suggested maximum time: 5 minutes)**

A motor company is introducing a new car in 2019. It will be available in 5 colours, 3 fuel types and 4 engine sizes, as shown.

| 2019 Car Range |             |             |
|----------------|-------------|-------------|
| Colour         | Fuel Type   | Engine Size |
| Red            | Diesel      | 1.3         |
| Blue           | Petrol      | 1.5         |
| White          | Electricity | 1.7         |
| Black          |             | 1.9         |
| Silver         |             |             |

- (a)** Calculate how many different types of this car can be bought.

[illegible]

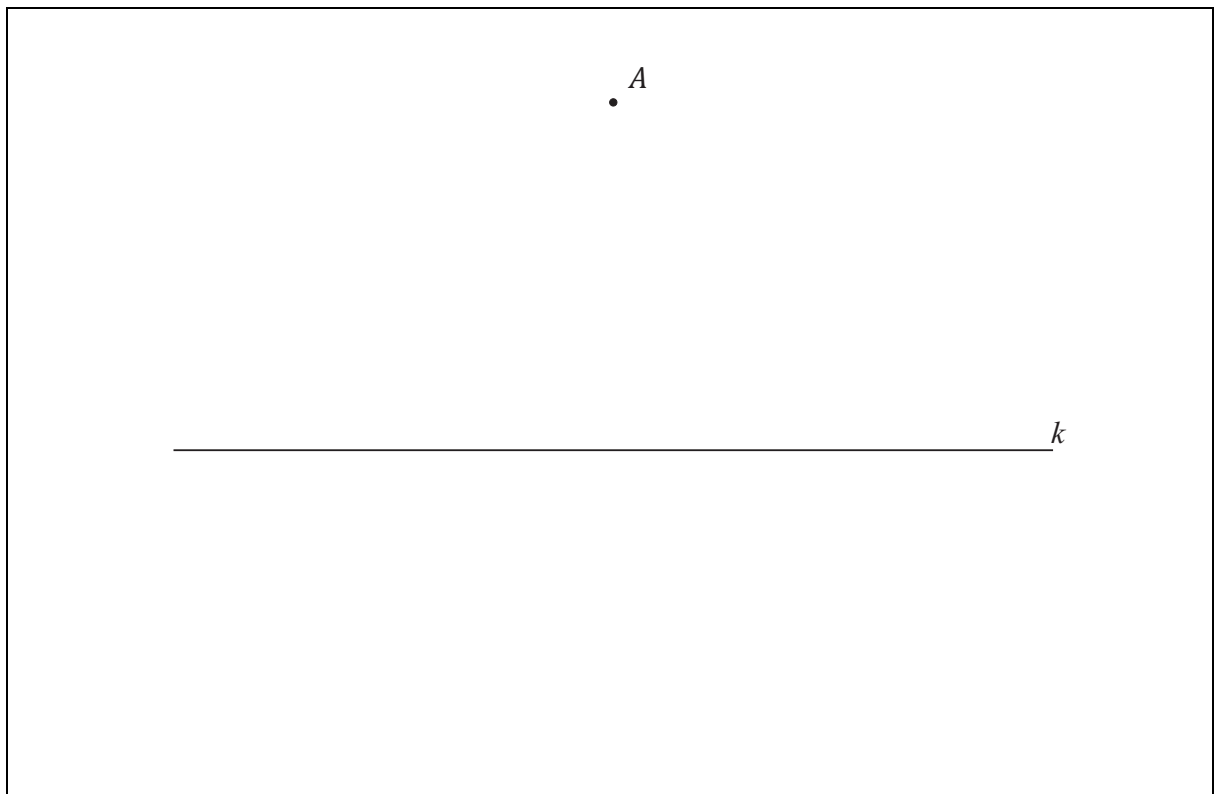
- (b)** The company wants to maximise the number of options a person has by increasing one of the options in a column above. Which column should it increase to give the greatest increase in options?

[illegible]

### Question 9

**(Suggested maximum time: 5 minutes)**

Using only a compass and a straight edge, construct a line perpendicular to the line  $k$  through the point  $A$  below. Show all of your construction lines clearly.



### Question 10

**(Suggested maximum time: 5 minutes)**

Prove that each exterior angle of a triangle is equal to the sum of the interior opposite angles.

*Diagram:*

|                      |  |
|----------------------|--|
| <i>Given:</i>        |  |
| <i>To Prove:</i>     |  |
| <i>Construction:</i> |  |
| <i>Proof:</i>        |  |

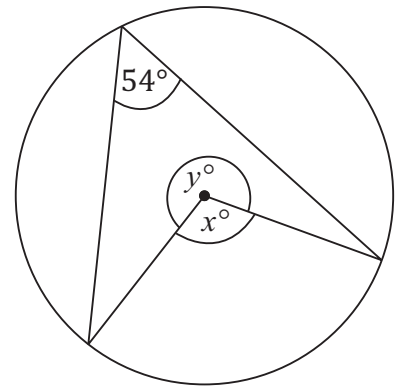
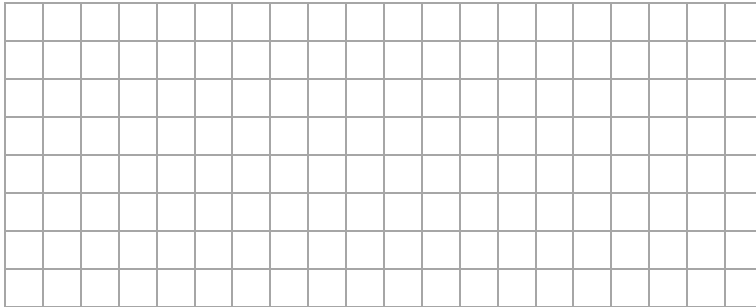
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### Question 11

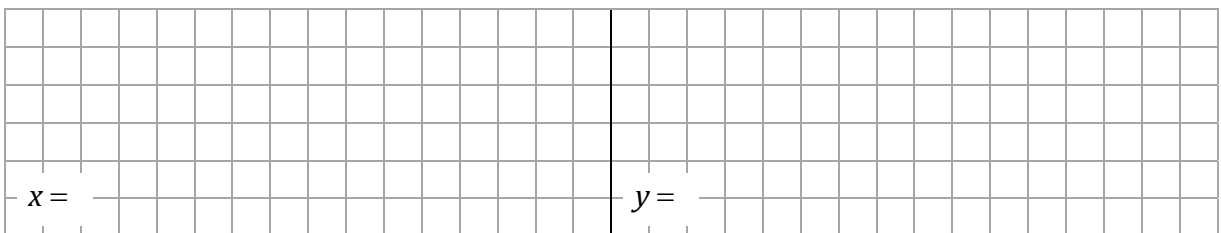
(Suggested maximum time: 15 minutes)

- (a) The area of the circle in the diagram is  $121\pi\text{cm}^2$ .

- (i) Find the diameter of the circle.

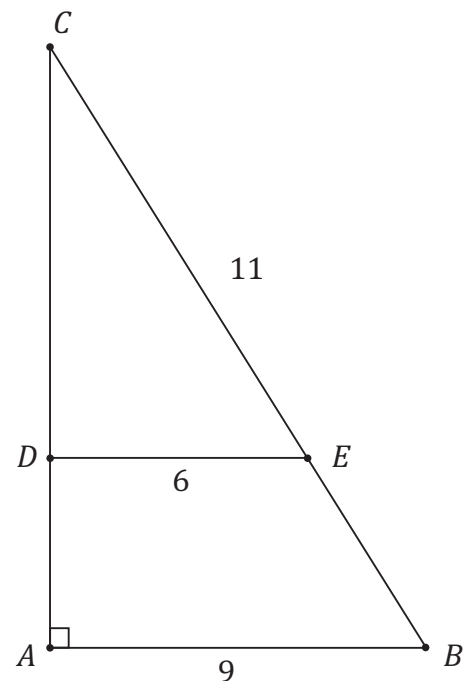


- (ii) Find the values of  $x$  and  $y$  in the diagram.

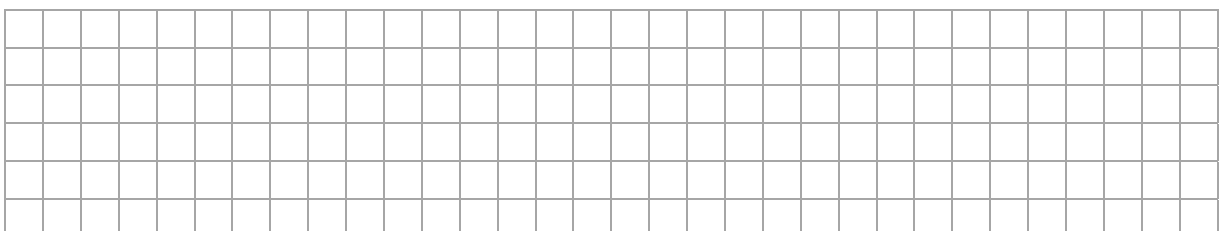


- (b) Right-angled triangles  $ABC$  and  $CDE$  are similar.  
 $|CE| = 11$ ,  $|DE| = 6$  and  $|AB| = 9$ , as shown.

- (i) Find  $|BE|$ .



- (ii) Using trigonometric ratios, find  $|\angle ABC|$ .  
 Give your answer correct to one decimal place.

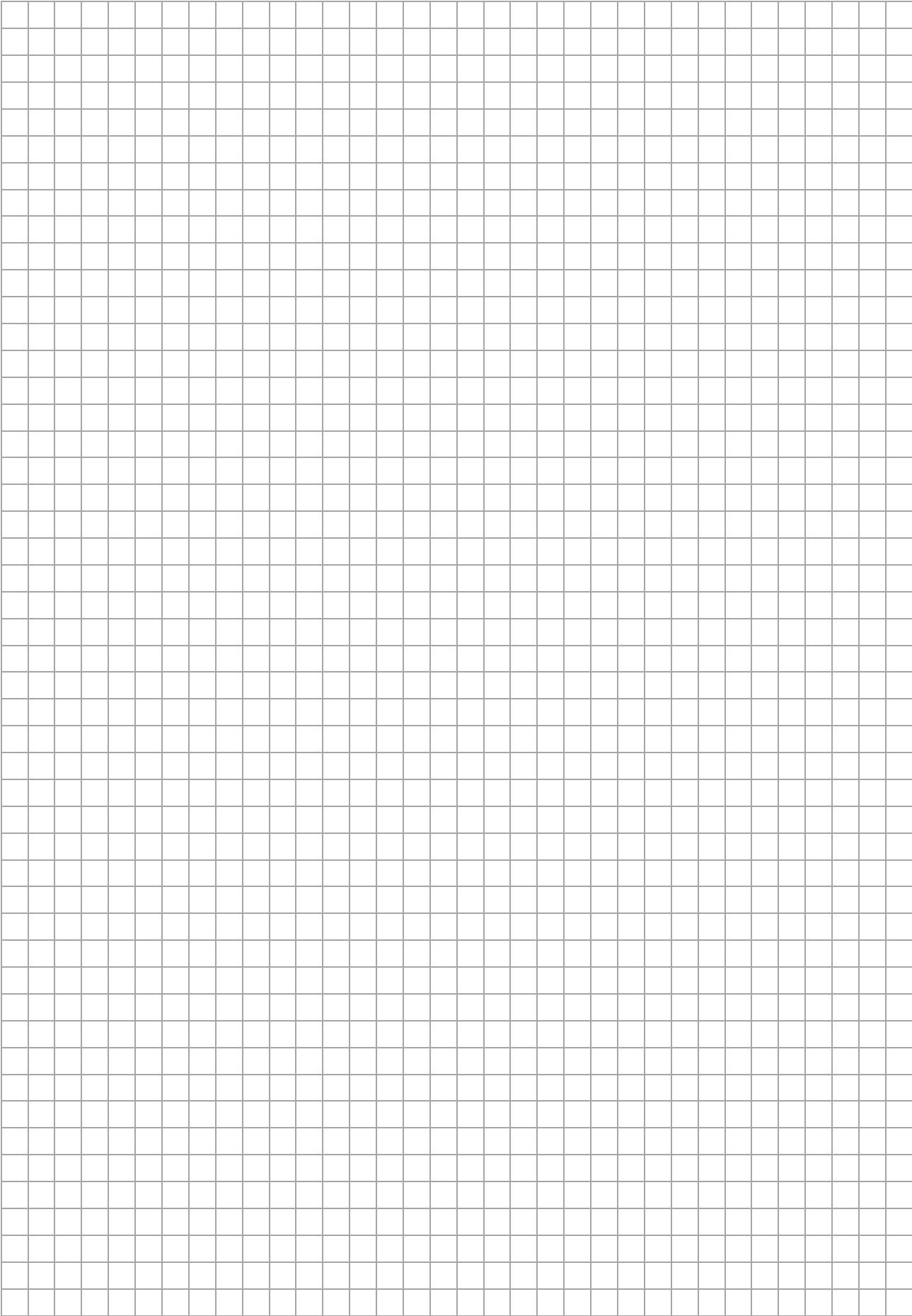




[illegible]

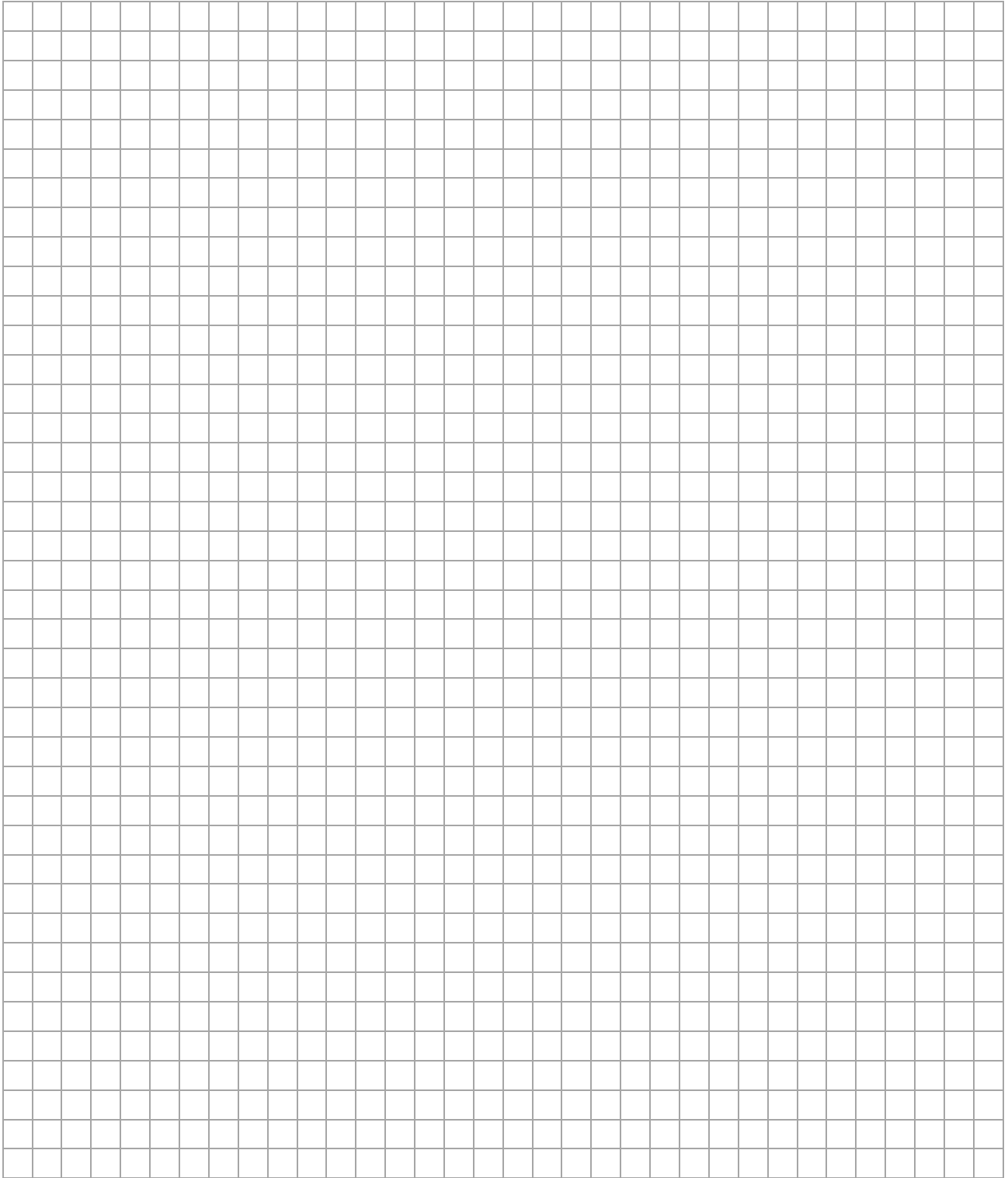
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*Mathematics*  
2018.1 J.20 17/20

You may use this page for extra work.



[illegible]

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Pre-Junior Certificate, 2018 – Higher Level

## Mathematics – Paper 2

Time: 2 hours, 30 minutes

